538. Convert BST to Greater Sum Tree

PLATFORM	DIFFICULTY	DATE
Platform: LeetCode	MEDIUM	Solved: August 3, 202

Problem Summary

Given the root of a BST, transform every node's value to the sum of all values greater than or equal to that node's value. Return the modified tree.

Approach

Reverse in-order traversal visits nodes in descending order. Maintain a running sum — add each node's original value to sum, then update node's value to the running sum.

Algo

1. nitialize sum = 0.
2. If root == null return null.
3. Recurse right first.
4. sum += root.val — accumulate.
5. root.val = sum — update node.
6. Recurse left.
7. Return root.

Key Insights

- Reverse in-order = descending order in BST — largest node visited first, sum accumulates correctly.
- Running sum at any node = sum of all nodes visited so far = sum of all values ≥ current node.
- Modifies tree in-place — no extra data structure needed.
- Same global variable pattern as LC 230 (kth smallest).

Time Complexity & Space Complexity

O(n) — every node visited once.

O(h) — recursion stack.

Approaches

- **Brute Force O(n²)**: For each node, traverse entire tree summing values ≥ current. Update node. O(n²) time.
- **Better O(n) space)**: In-order traversal → store in array → compute suffix sums → map back to nodes.
- **Optimal O(1) space**: Reverse in-order (right → root → left) with running sum — visits nodes largest to smallest, accumulates sum on the fly.